Tristan Lecourtois

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Website
LinkedIn
GitHub

Education

2025–2026 **Master of Science in Mathematics, Vision, Learning (MVA)**
École Normale Supérieure Paris-Saclay
Machine Learning, Deep Learning, Optimization, Reinforcement Learning, NLP

2023–2026 **Master of Engineering**
Mines de Saint-Étienne, France
Computer Science (Machine Learning, Python, C, C++, Algorithmics, SQL)

2021–2023 **Classes Préparatoires (MPSI/MP)**
Lycée Pierre Corneille, Rouen
Equivalent to a Bachelor's degree in Mathematics and Physics

Research Experience

Mar. 2026 – June 2026 **Research Intern**.
InstaDeep, Paris

Pre-trained a multimodal foundation model from scratch in oncology (H&E histopathology, RNA-seq, DNA) via large-scale self-supervised masked modeling with distributed multi-GPU training. Built evaluation and benchmarking pipelines against state-of-the-art foundation-model baselines on downstream clinical tasks.

Apr. 2025 – Aug. 2025 **Visiting Student Researcher**.
NASA Jet Propulsion Laboratory (JPL), Los Angeles, CA

Extended ROSA, an LLM/VLM-powered agent for natural language control of robots via a reasoning-action-observation loop. Prototyped episodic memory for contextual recall, and a DINOv2 + SAM vision pipeline for grounding-based object detection and segmentation.

Jan. 2024 – June 2024 **Research Intern**.
Atos, Aix-en-Provence

Fine-tuned open-source foundation models (LLMs) for Text-to-SQL, reaching 90% query execution accuracy. Built a Retrieval-Augmented Generation (RAG) framework with LangChain and Ollama for conversational access to structured and unstructured data.

Research Projects

June 2026 **Drug-Perturbation JEPA World-Model** [Code] [Slides]
World Model Hackathon
Built an energy-based JEPA world-model that predicts how a cell's transcriptomic state changes under a drug intervention, learning the perturbation entirely in latent space on the Tahoe-100M single-cell dataset. Extended eb-JEPA with a drug-conditioned predictor, anti-collapse regularization, and biology-informed losses, enabling in-silico drug screening.

Dec. 2025 **Cross-Modal Alignment for Molecular Graph Captioning** [Code] [Report]

ALTeGraD Course, ENS Paris-Saclay

Built a hybrid cross-modal framework (MolCA-based) translating molecular graphs into captions, aligning a GINE graph encoder with a Galactica-1.3B LLM via a Q-Former projector. Benchmarked three approaches: contrastive dual-encoder retrieval (SciBERT/RoBERTa), GNN-to-T5 generation, and soft-prompting.

Oct. 2024 **Early Detection of Endometriosis from Patient-Reported Symptoms** [Code]

Personal Research Project

Built a non-invasive ML screening tool predicting endometriosis from patient-reported symptoms (800+ entries). Used the Jaccard Index for feature selection (56 to 24 symptoms) and benchmarked ML algorithms with 10-fold cross-validation, yielding an interpretable symptom ranking.

Technical Strengths

Languages Python, C++, LaTeX, HTML, CSS

Frameworks PyTorch, TensorFlow, NumPy, OpenCV, Pandas, scikit-learn, ROS

Tools Linux, Git, Docker

Relevant Coursework

ENS Paris-Saclay Representation Learning for Computer Vision and Medical Imaging, Reinforcement Learning, Machine Learning for Time Series, Algorithms for speech and NLP , Geometric Data Analysis, Advanced Learning for Text and Graph Data, Generative Modeling for images

Awards and Achievements

2026 **Mistral AI Worldwide Hackathon (Top 100 / 7,000+ applicants)** – Developed AI-powered 911 emergency dispatch system *(Code & Demo)*

2026 **Participant at HackEurope, the largest student hackathon in Europe** – Developed Feelow, a personal finance Agent analyzing Polymarket prediction markets to generate trading insights, combining LLM-powered search, sentiment analysis, and quantitative scoring *(Code & Demo)*

2025 **Recipient of the Fondation Mines-Télécom International Scholarship** for research at NASA JPL

2025 **Participant at UC Berkeley AI Hackathon** – Developed an AI Therapist project *(Project Link)*

2025 **Y Combinator AI Startup School** – Selected participant for the intensive AI entrepreneurship program